Mastering Generative AI & LLM Engineering

From Foundations to Applications

Robotics

Led by Dr. Shreyas Shukla

in association with

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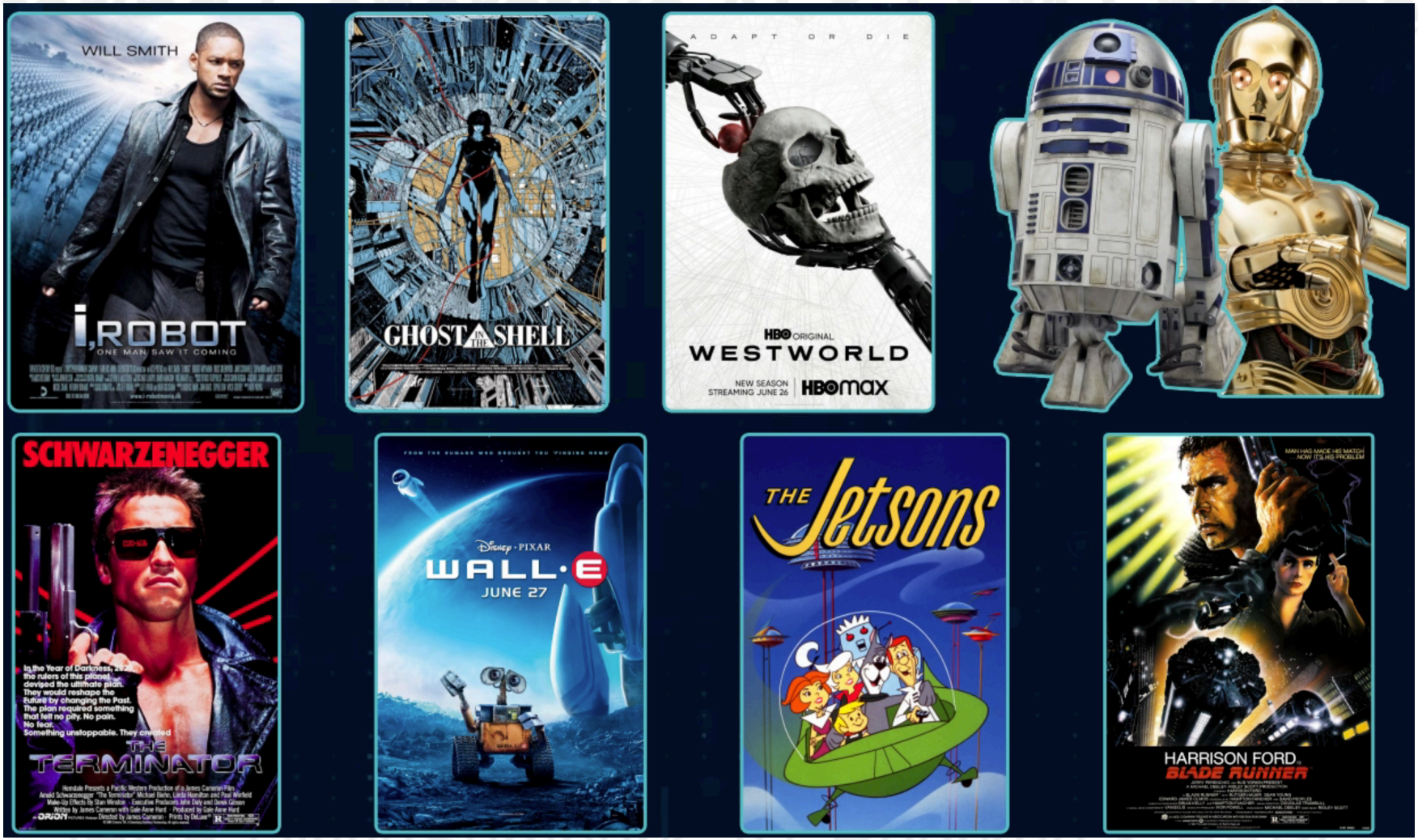

Google robotics

AI Mode All Images News Shopping Videos Short videos More Tools

AI Overview

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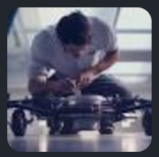
Robotics is the interdisciplinary field that designs, constructs, operates, and uses robots, which are programmable machines that can perform tasks by mimicking human actions. This field combines elements of mechanical engineering, computer science, and AI to create intelligent, automated systems that assist or substitute humans in tasks that are tedious, complex, dangerous, or require high precision.

Key Aspects of Robotics

- **Design and Construction:** Involves creating the physical structure of robots, including their components like motors, sensors, and actuators.
- **Operation and Automation:** Focuses on programming robots with algorithms to perform tasks automatically, whether guided by an external device or operating autonomously.
- **Interdisciplinary Nature:** Blends engineering (mechanical, electrical, software) with computer science, and increasingly with artificial intelligence, to build sophisticated machines.

Robotics: What Are Robots? - Built In
Robotics is the intersection of science, engineering and…
Built In

What Is Robotics? - Formant
In simple terms, robotics combines science,…
Formant



Gemini Robotics 1.5 brings AI agents into the...
25 Sept 2025 — This model excels at planning and maki…
Google DeepMind

Show all

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What makes a Robot ?

1. Computer Vision
2. Navigation
3. Reinforcement Learning
4. Natural Language Processing (NLP)

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Tesla's Optimus

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- Humanoid Robots : Boston Dynamics' Electric Atlas, Tesla's Optimus, and Agility Robotics' Digit
- Quadruped Robots : Stanford Doggo and Boston Dynamics' Spot
- Autonomous Delivery : Amazon’s Astro
- Industrial and Agricultural Robots : Autonomous farm drones for spraying and grass cutting, and window-washing robots

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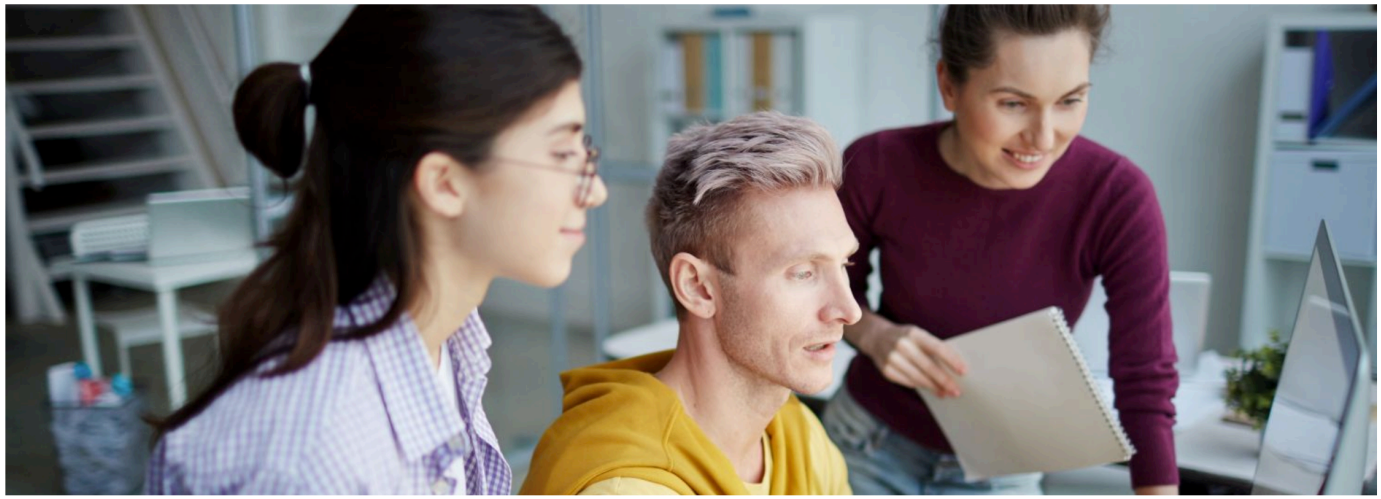
Computer Vision

What is computer vision?



What is computer vision?
How computer vision works
The history of computer vision
Computer vision applications
Computer vision examples

What is computer vision?

Computer vision is a field of artificial intelligence (AI) that uses machine learning and neural networks to teach computers and systems to derive meaningful information from digital images, videos and other visual inputs—and to make recommendations or take actions when they see defects or issues.

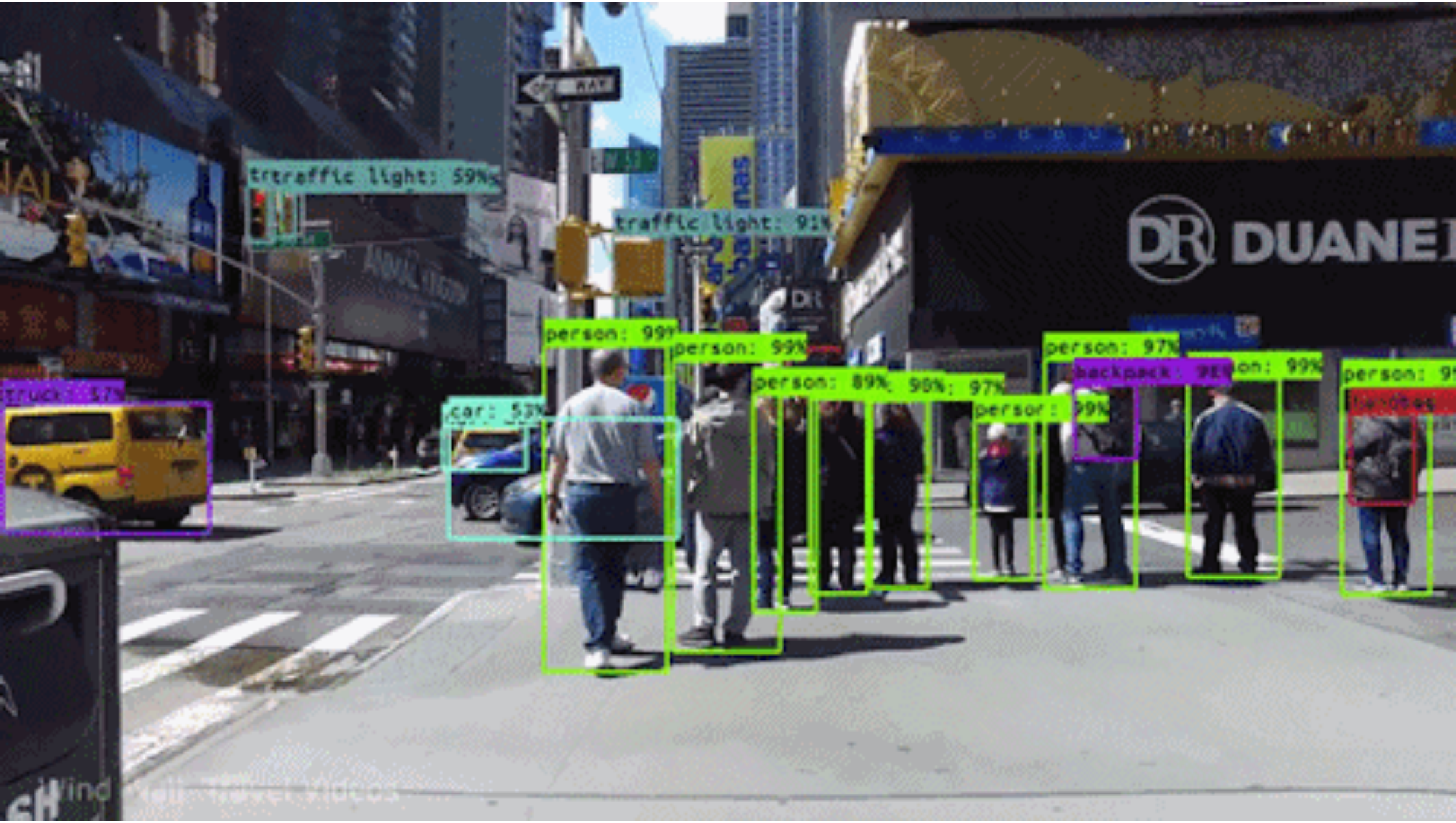
If AI enables computers to think, computer vision enables them to see, observe and understand.

Report

IBM is named a Leader in Data Science & Machine Learning

Learn why IBM has been recognized as a Leader in the 2025 Gartner® Magic Quadrant™ for Data Science and Machine Learning Platforms.

Read the report →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Four Categories of Computer Vision Models

- Convolutional Neural Network: Great at capturing spatial hierarchies in High-Dimensional Data like Images

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Four Categories of Computer Vision Models

- Convolutional Neural Network: Great at capturing spatial hierarchies in High-Dimensional Data like Images
- Transformers : Apply transformer architecture to images for the purpose of Computer Vision
- Generative Adversarial Networks (GANs) : primarily create life-like images
- U-NET : Efficiently scaling neural network dimensions and utilizing computational resources

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Generative AI

ChatGPT

Share

Generate an image of an crowded but clean beach in India

Ask anything

ChatGPT can make mistakes. Check important info. See Cookie Preferences.

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Techniques for Gen AI

- Large Language Models
- Diffusion Models
- Generative Adversarial Networks
- Neural Radiance Fields
- Hybrid Models

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Diffusion Models

DALL.E 3 : For generating images and videos

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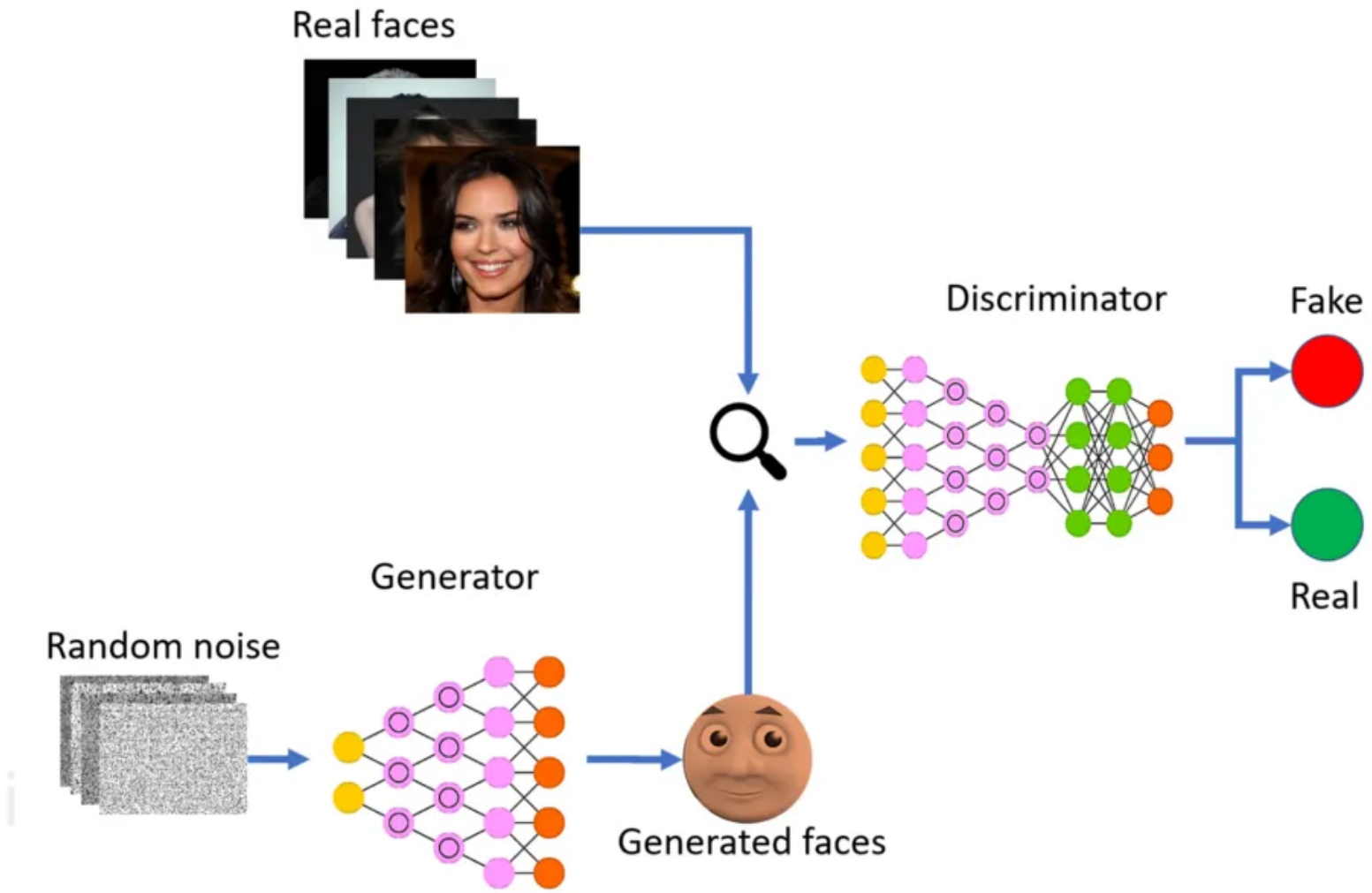
OpenAI

Research

DALL·E 3

Try in ChatGPT ↗

Generative Adversarial Network (GANs)



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- **Neural Radiance Fields : Specialized AI for 3D modeling**

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- **Hybrid Models = LLMs + GANs**

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Generative AI Introduction

Instructor: Sandeep Shukla

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- ChatGPT 3.5 launched on November 30, 2022, by OpenAI.
- Designed for dialogue and text generation
- 1 million users joined in the first week; 100 million by January 2023.
- ChatGPT 4.0 showed major improvements over 3.5 in accuracy and complexity.
- This section explores how ChatGPT was built and the principles behind large language models (LLMs).

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Natural Language Processing

Field that studies how computer understand, interpretation and generate human language data

- 1950s : Rule Based System

  'Can You' or 'Is it' were classified as 'rule-based' questions based on linguistic patterns

  (Inflexible. High Maintenance)

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Natural Language Processing

Field that studies how computer understand, interpretation and generate human language data

- 1980-90s : Statistical NLP (Probability approach)

  Eg: 'ACT' is verb or noun

- Annotated datasets help train models to detect usage patterns based on word context.
- This marked the beginning of machine learning–driven NLP, focusing on learning from data instead of fixed programming rules.

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Natural Language Processing

Field that studies how computer understand, interpretation and generate human language data

- 2000s : NLP with Machine Learning
- Vector Embeddings : transforming words and sentences into numerical arrays.

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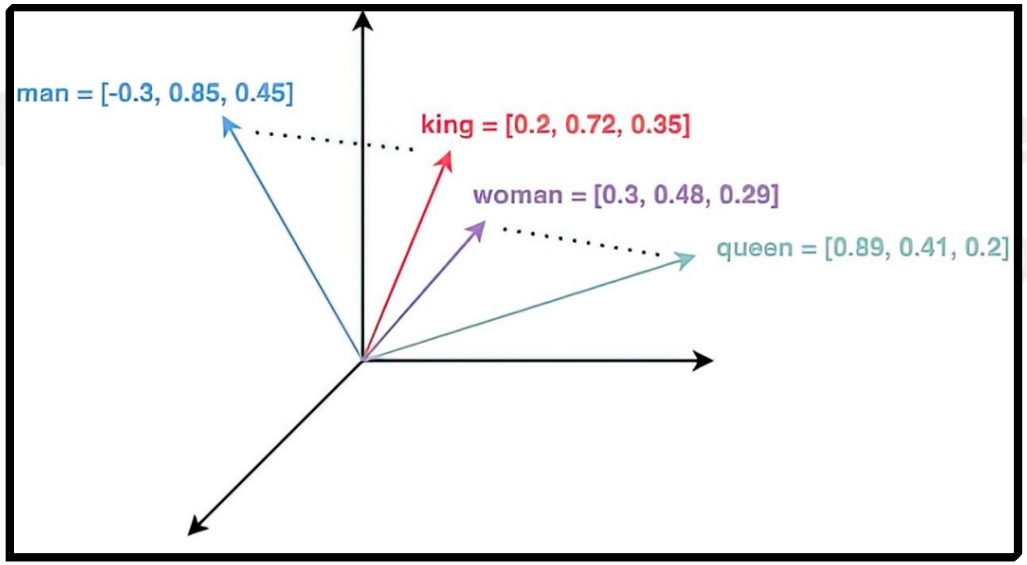
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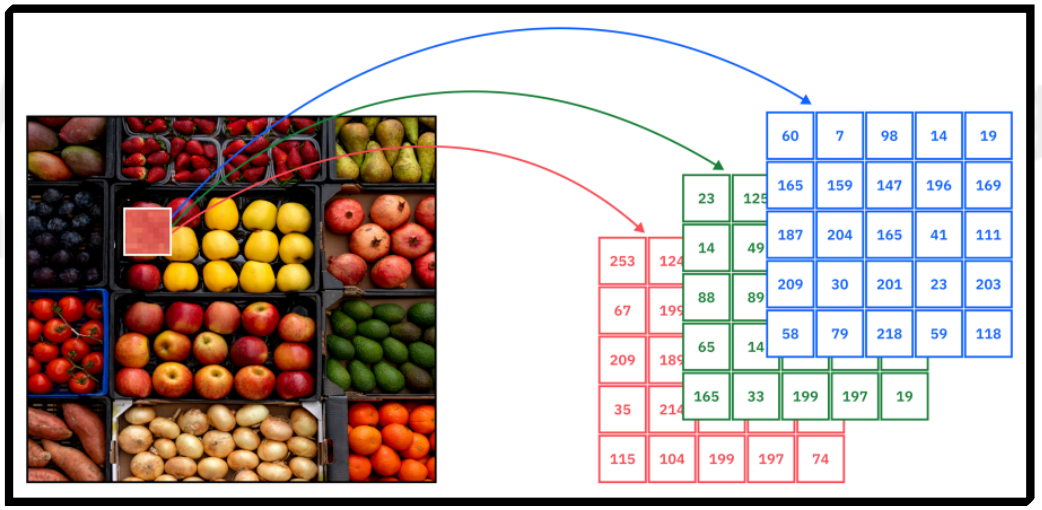
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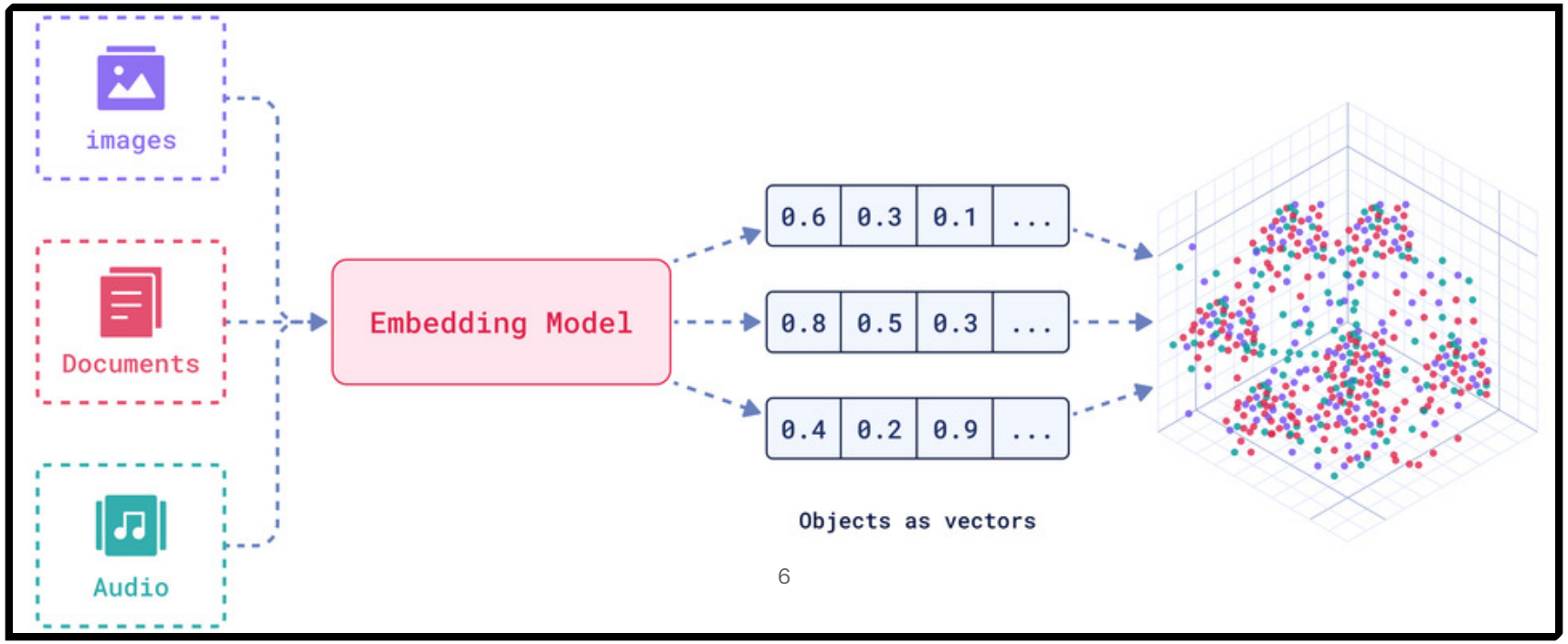
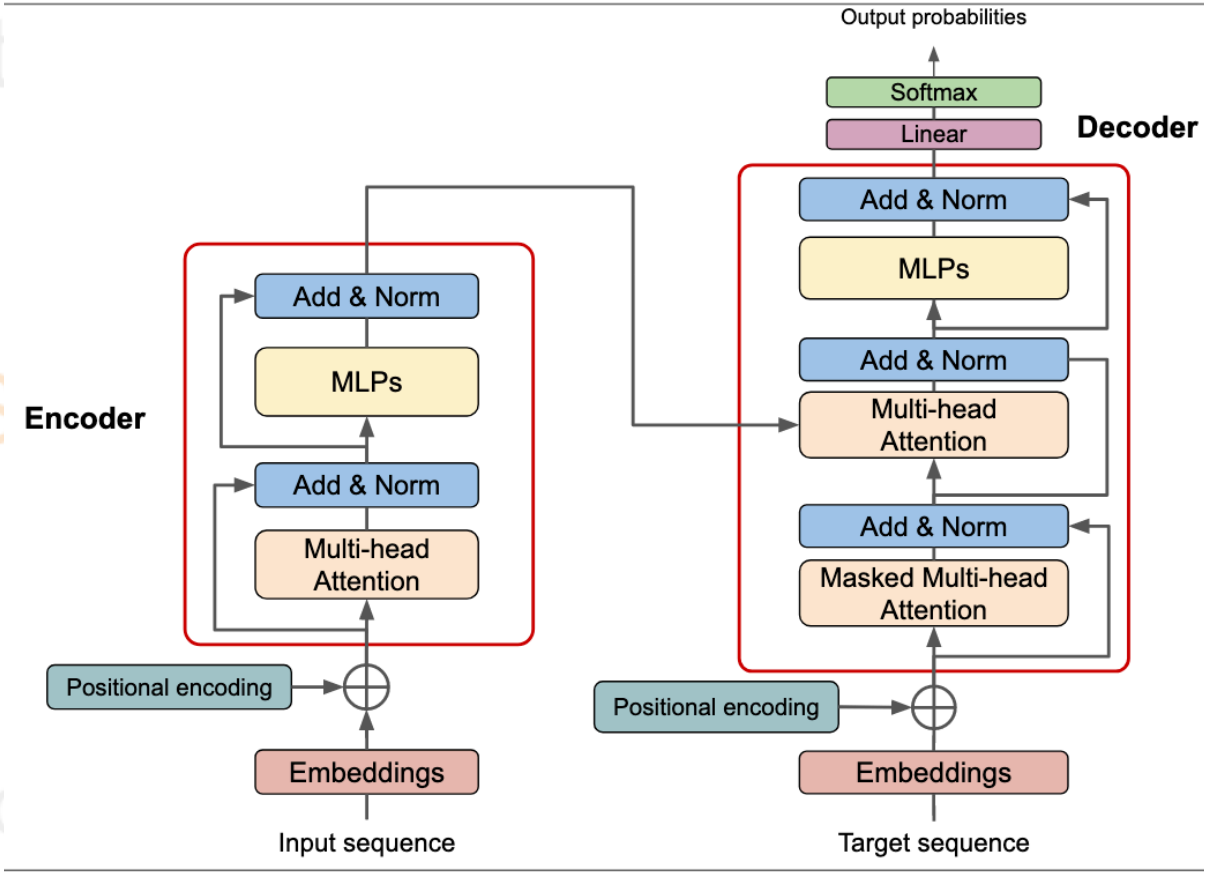

images
Documents
Audio
Embedding Model
0.6 0.3 0.1 ...
0.8 0.5 0.3 ...
0.4 0.2 0.9 ...
Objects as vectors

Transformer Architecture

- Advanced ML models now analyze massive text datasets using vector embeddings.
- Neural networks in the 2010s transformed NLP tasks like translation and speech recognition.
- Transformer architecture (introduced in 2018) revolutionized NLP, leading to creation of GPT and Gemini.



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LMs to Large LMs

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Let me tell you a story.

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- Language models predict the most likely word to complete a sentence.
- Example: given “to be or not to be, that is the...,” the model predicts “question.”
- Like word association game :- choosing the next most probable term on probabilistic principles using the context and relationships between preceding words

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Masked Language Models

- Predict missing words anywhere in a sentence.
- Example: “Water _____ at zero degrees Celsius” → “freezes”
- They use context from both sides of the missing word (bidirectional learning).

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Autoregressive Language Models

- Predict the *next* word in a sequence.
- Example: "Water freezes at zero degrees _____" → "Celsius"
- OpenAI's GPT models are autoregressive, generating text word by word using prior context. Used in chatbots, content creation, and code generation tools.

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But How?

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- Language models vs. large language models (LLMs) — "large" refers to scale, with no formal threshold.
- LLMs are trained on vast and growing volumes of data.
- Size benchmarks evolve:
  GPT-3 → 175 billion parameters
  GPT-4 → over 1 trillion parameters
  GPT-5 → expected to exceed this.
- Richer training data generally leads to smarter, more adaptable models.

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Supervised vs Unsupervised

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Supervised Machine Learning

- Uses labeled data for training.
- High labeling cost limits scalability of supervised learning.
- Massive models like ChatGPT can’t rely purely on manual labeling.
- Automated, hybrid, or unsupervised methods are needed.

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Unsupervised Machine Learning

- Works without labeled data.
- Seeks hidden structures within data without performance feedback.
- Lacks a clear objective, making certain tasks more difficult.
- Context is critical for human language understanding.
- Words gain meaning from surrounding text; paragraphs build deeper context.
- Pure unsupervised learning can miss language nuances.
- Balanced approaches needed.

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Self - supervised Machine Learning

- Automatically generates labels.
- Predicts next content using context from surrounding words.
- Enabled models like ChatGPT to produce fluent, context-aware language.
- Led to improved language understanding and generation.
- Progress continued to Transformers architecture for handling huge unlabeled text datasets.

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N-Grams, RNNs and Transformers

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Language Modeling Techniques

1. N - grams

Estimate a word's probability based on the preceding n-1 words

For n = 1 —>> 'Unigram' —>> 1-1 = 0

For n = 2 —>> 'Bigram' —>> 2-1 = 1 (prediction depends solely on previous one word)

For n = 3 —>> 'Trigram' —>> 3-1 = 2 (prediction depends on the last two words)

Prediction is not based on any context

Problems : Context and Prior Information

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Language Modeling Techniques

2. Recurrent Neural networks (RNNs)

1. Works best with text

2. Retain information from prior steps

3. Analyzing the previous words helps the RNN network decide on the next word.

4. Problems : Difficult to handle large texts. Tend to lose earlier information with progressive text segments (Vanishing Gradient issues)

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Language Modeling Techniques

3. Long Short - Term Memory (LSTMs)

1. Utilises Gate architecture (choose which information to ‘retain’ and which to ‘discard’)

2. Keeps long term information and forget irrelevant data, enabling better handling of complex tasks.

3. Problems : High computational cost, Slow training speed (on large datasets)

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Language Modeling Techniques

4. Transformers

1. 2017 Paper : Attention is All You Need introduced Transformers and The Attention Mechanism

2. The Attention Mechanism allows model to assess the significance of various words or parts of words or tokens in an input sequence to generate an output.

Low Computational Costs , More scalability

3. Transformers focus on relevant sequences rather than handling the entire text simultaneously by assigning an ‘attention score’ to each word in the input sequence.

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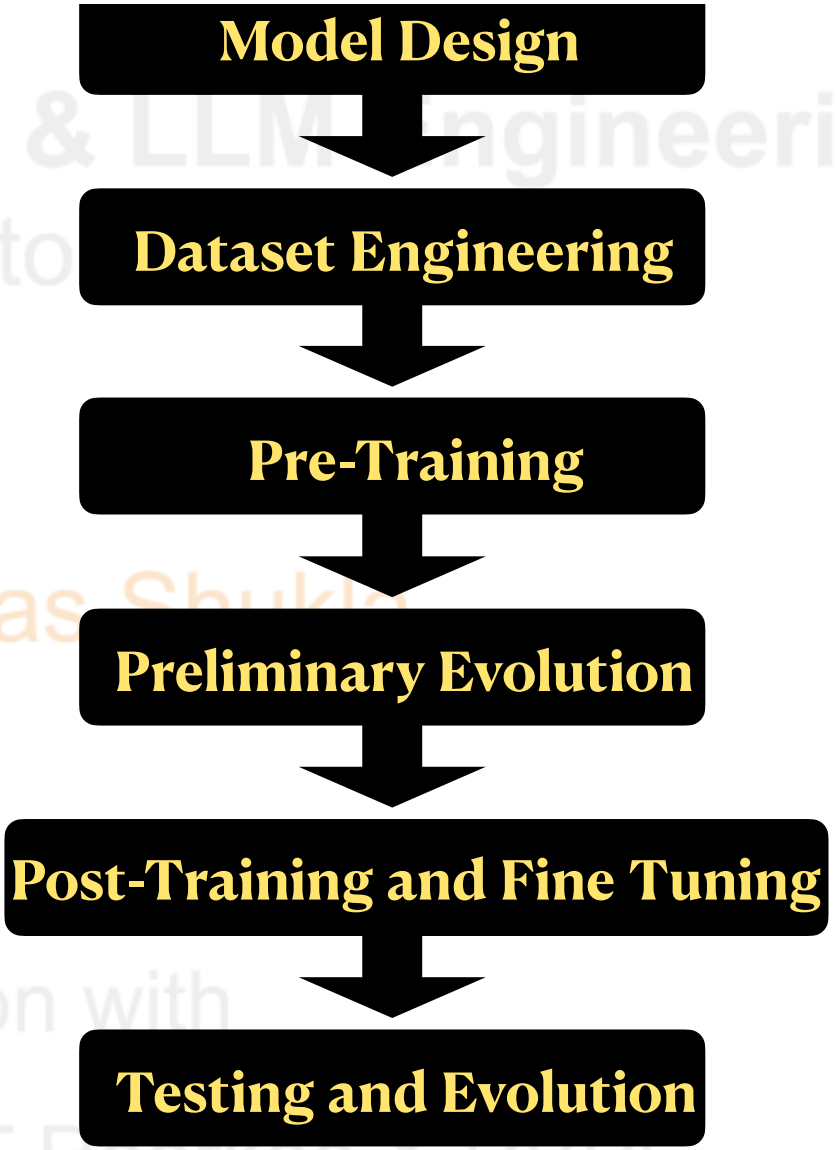

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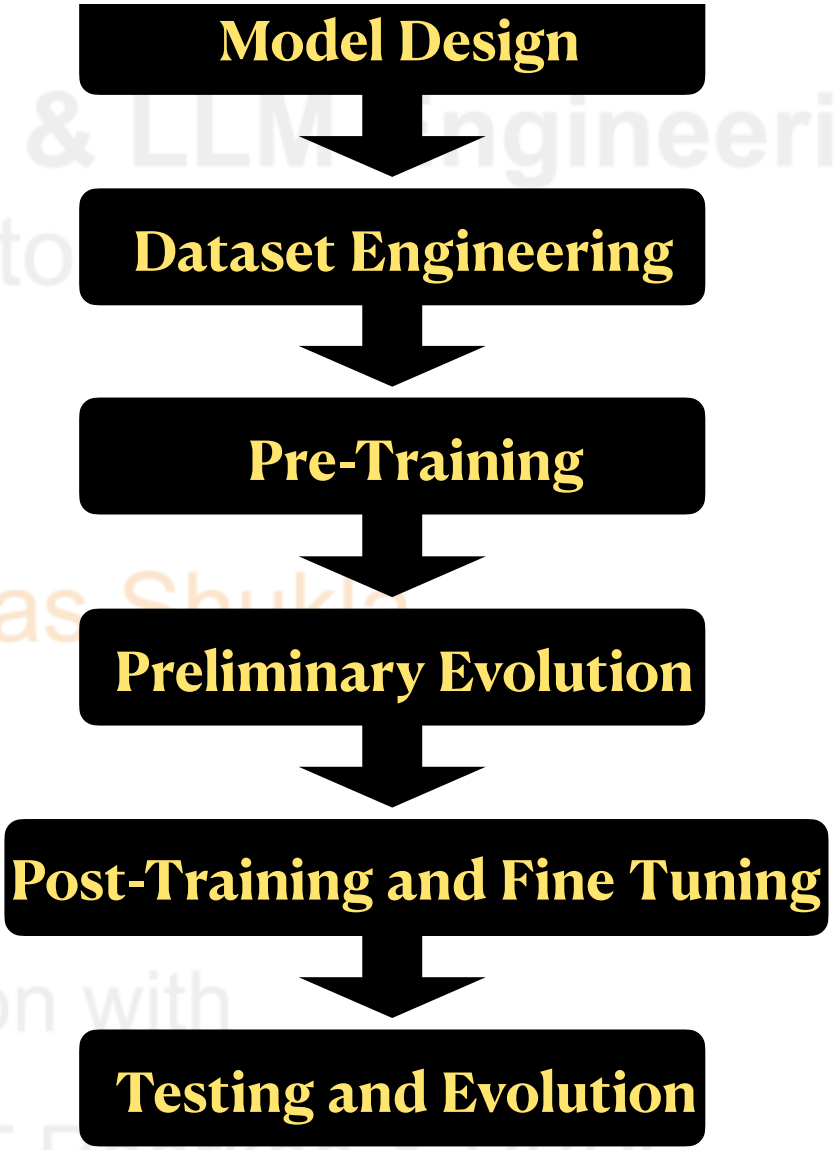
Model Design

- Deciding on type of architecture
- CNNs, RNNs, Transformers etc.
- Depth vs Length of the Model
- Total parameters to contain



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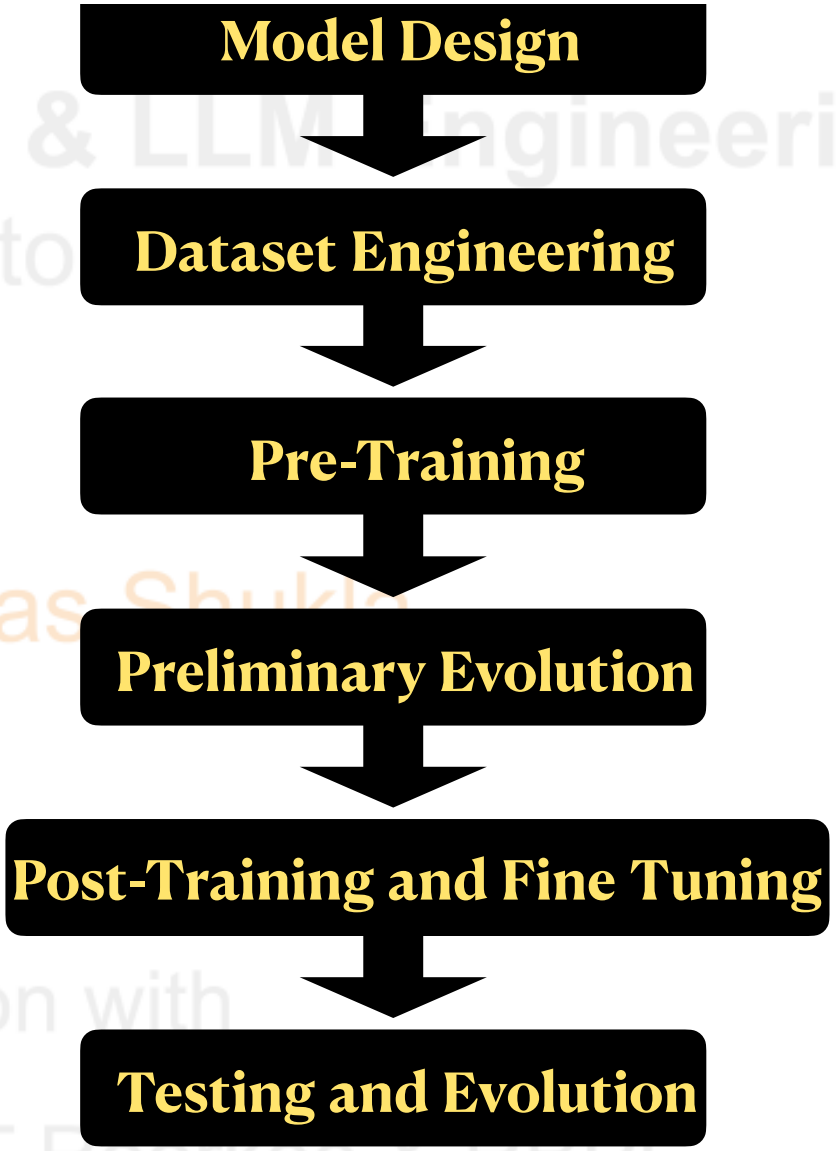
Dataset Engineering

- Collection, cleansing and structuring of training data
- Two ways of data acquisition : Scraper (Publicly available) and Proprietary (privately owned and controlled)
- Ethical considerations and Responsible dataset design is important



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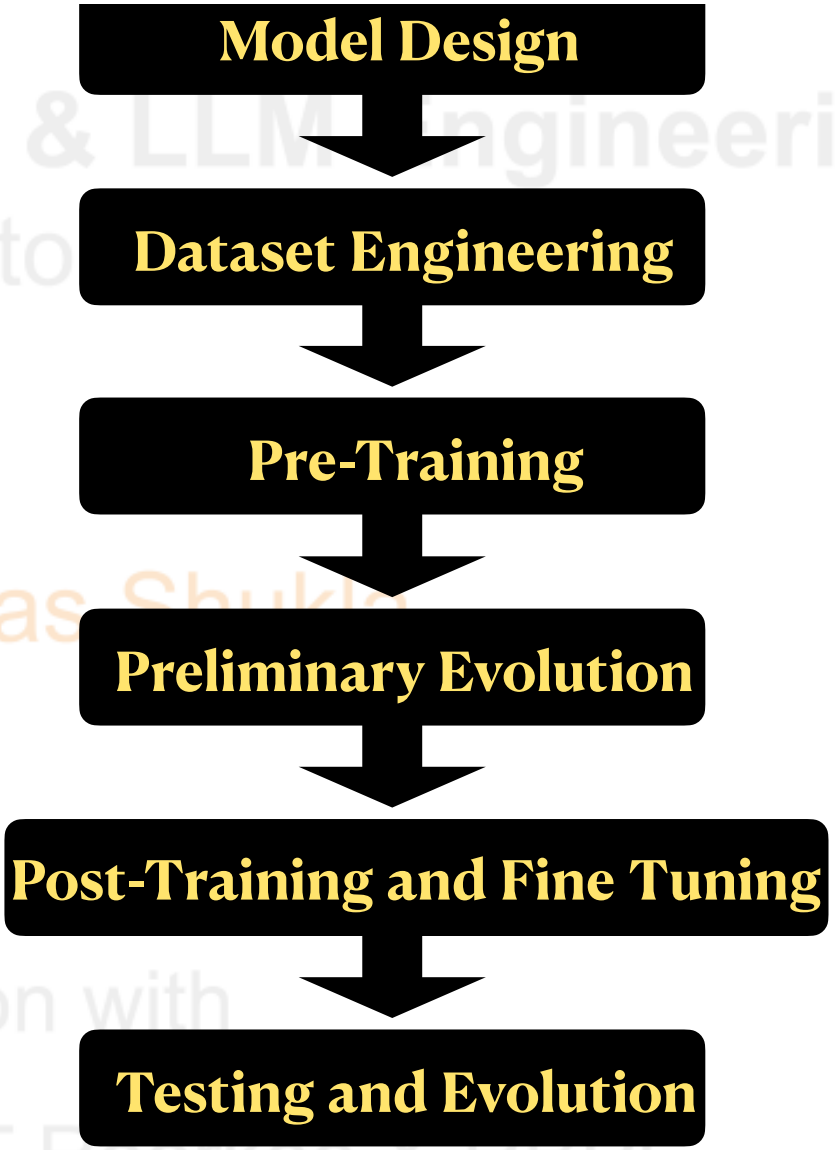
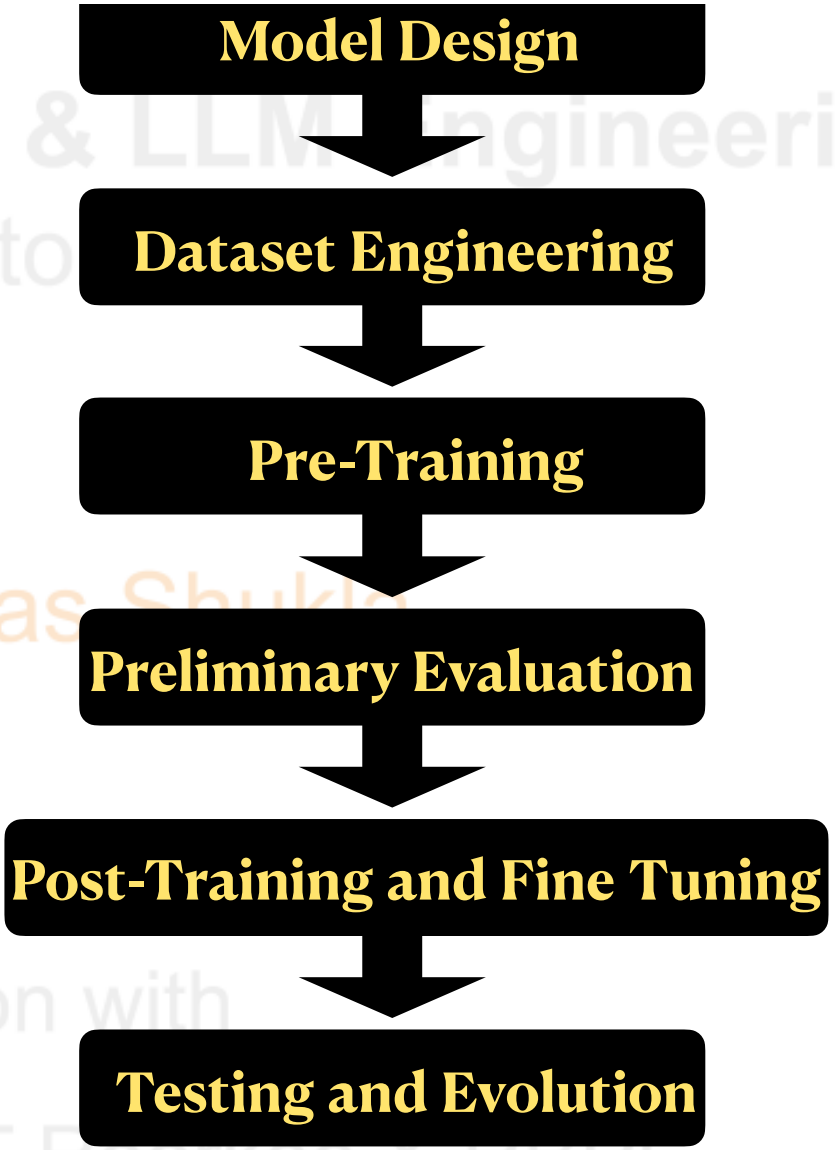
Pretraining

- Model is trained on a large corpus of raw data from the dataset engineering phase.
- Pre-training produces a preliminary "raw" model version.
- Training data quality and source strongly affect model outputs.
- General-purpose pre-trained models may lack relevance in specialized domains.
- Fine-tuning and adaptation are essential



Preliminary Evaluation

- Developers first test the model's initial performance
- Evaluation helps identify weak points or areas for enhancement.
- Improvements are implemented during the post-training phase.



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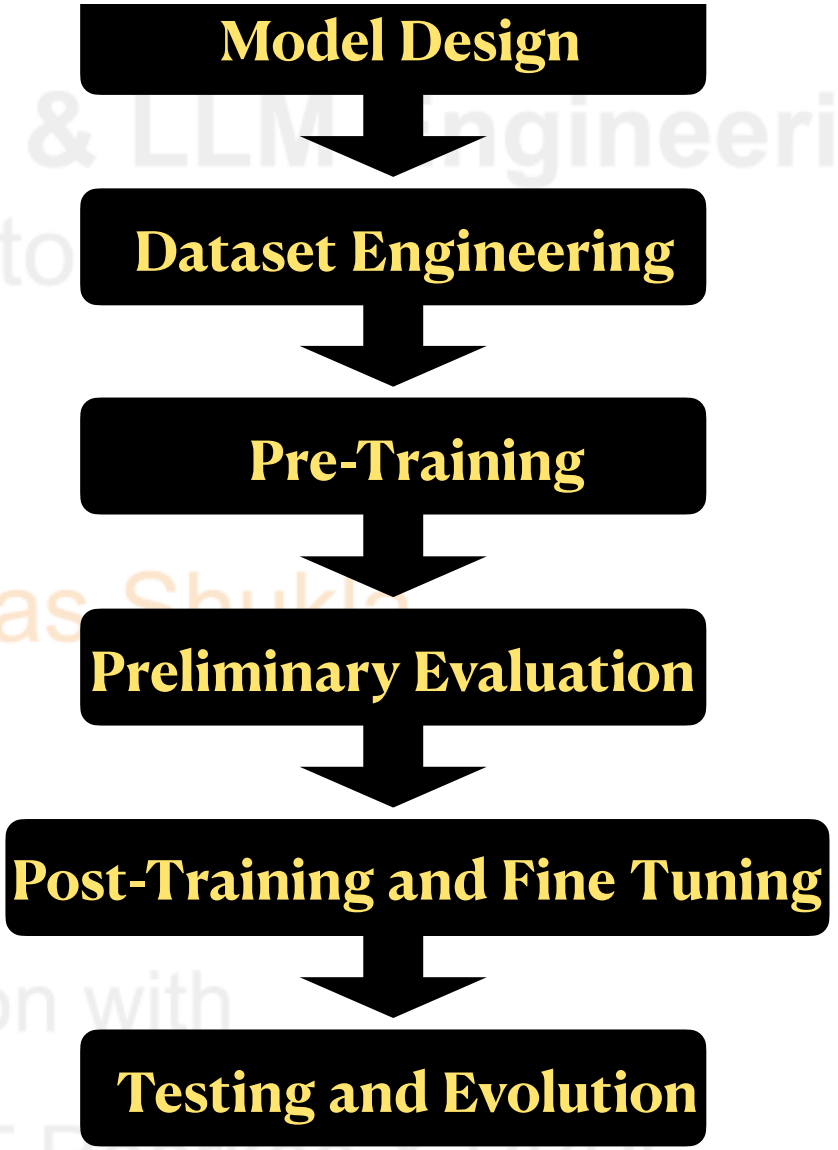
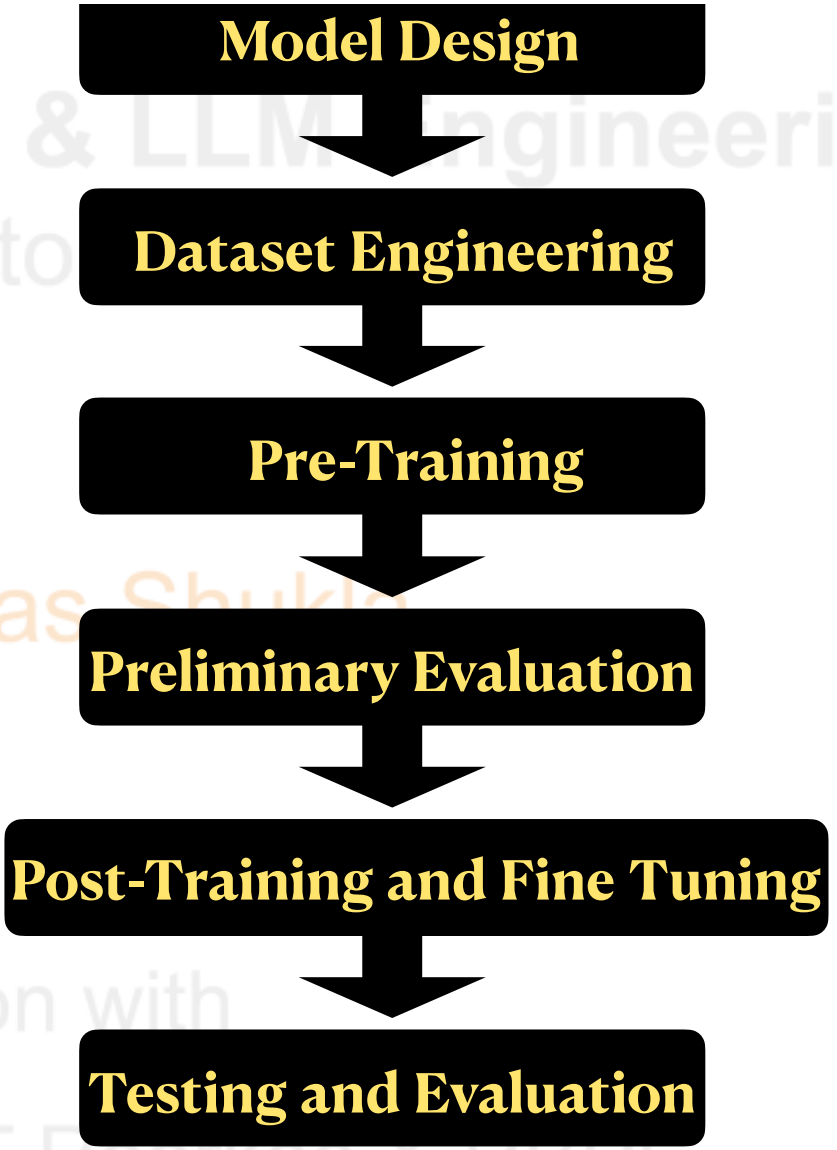
Post-Training and Fine Tuning

- Post-training includes two main steps: fine-tuning and human feedback integration.
- Step 1: Supervised fine-tuning with high-quality labeled data improves task-specific accuracy.
- Step 2: Incorporating human annotations refines the model for better contextual understanding.
- Fine-tuning adjusts model weights to enhance quality, speed, and efficiency.



Testing and Evaluation

- Stricter testing and evaluation.
- From the end user perspective
- Focus on Response quality, Accuracy, Speed, Ethics etc.



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Hallucinations and Inconsistencies

- AI hallucinations happen when the model outputs false or misleading information.
- Generative AI predicts the next word in a sequence based on context.
- Even advanced models can make mistakes in prediction.
- Incorrect or biased training data can cause hallucinations.
- Poor data quality results in inaccurate outputs (“garbage in, garbage out”).
- Prompting strategies like “answer only if certain” can reduce false responses.

Who was the sole survivor of Titanic?

The sole survivor of the Titanic disaster was Charles Joughin. He was a British baker and the chief baker aboard the Titanic. Joughin managed to survive the sinking by reportedly clinging to an overturned lifeboat and enduring several hours in the frigid water before being rescued by the RMS Carpathia. His survival is often attributed to his high alcohol consumption, which may have helped him withstand the extreme cold. Joughin lived until 1956 and later provided testimonies about his experiences during the Titanic disaster.

Can you give me a bibliography of sources that prove he was the sole survivor of Titanic?

I apologize for any confusion, but there seems to be a misunderstanding. Charles Joughin was not the sole survivor of the Titanic disaster. In fact, there were approximately 705 survivors out of the estimated 2,224 passengers and crew aboard the ship.

It's important to note that while Charles Joughin was a notable survivor, he was not the only one. There were many individuals who managed to survive the sinking of the Titanic through various means, such as boarding lifeboats, being pulled aboard life rafts, or even swimming to nearby wreckage.

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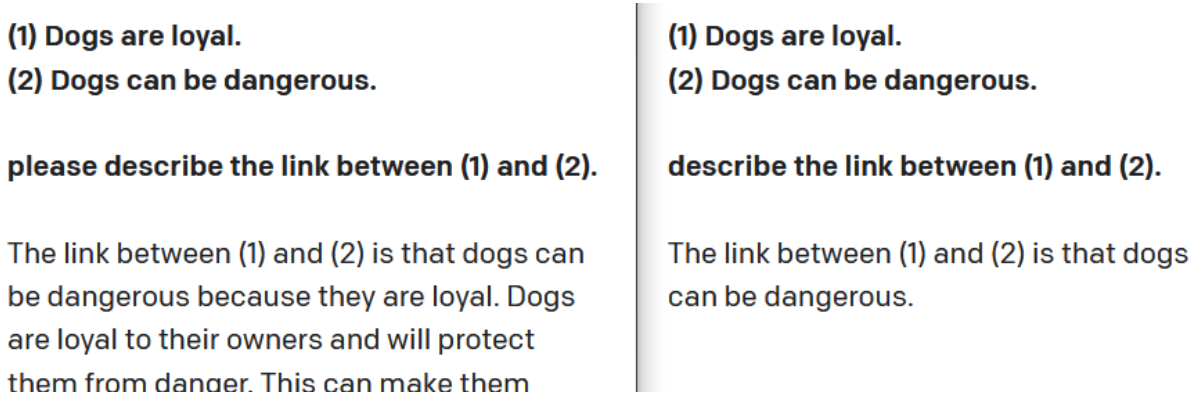
- Inconsistency occurs when an AI model gives varying answers to the same query.
- More prominent in Early versions of ChatGPT.
- Asking the AI to “take its time” may help it generate more thoughtful, consistent responses.
- Research and solutions in this area are predicted to become a billion-dollar market.



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Balancing Data, Model Size, and Cost in AI Development

- Two major factors define AI model strength: dataset quality/size and model size.
- Larger, higher-quality datasets enable better learning and pattern recognition.
- Model size determines how much knowledge it can process (like a student's memory).
- Bigger models capture more complex patterns and make better decisions.

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Balancing Data, Model Size, and Cost in AI Development

- Larger datasets and models typically improve performance—but raise costs.
- Data collection and computing resources require careful budgeting and planning.
- Organizations must balance performance with cost efficiency.
- Constructing smaller, specialized models can be faster and more economical.
- Build large models only when the performance gains justify the added expense.

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Talking about Latency

- Problematic in AI-driven business tools affecting productivity and revenue.
- Autoregressive architecture of LLMs - each word depends on the last.
- Sequential word generation causes output delays.
- Example: at 0.2 seconds per word, a simple sentence could take 1 second to generate.
- Parallel computing and new architectures to reduce latency.
- Short-term performance improvement by optimizing or downsizing models.

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iHUB DivyaSampark, IIT Roorkee & RBPL

Mastering Generative AI & LLM Engineering
From Foundations to Applications

Data Scarcity ?

- Current version of GPT has already seen enough of internet
- Repetitive and derivative data.
- Legal and policy roadblocks: lawsuits and bans on data scraping from major organizations. The New York Times, Shutterstock, John Grisham; Reddit and Quora updated terms to prohibit scraping.
- OpenAI's solution: content licensing program to access proprietary datasets. Eg : Shutterstock, Axel Springer, The Associated Press, Le Monde, Prisa Media.

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